CSA0307 - Data Structures

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Analytical Problem

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1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack.
Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow.

Feature	Recursion (Algorithm A)	Explicit Stack (Algorithm B)
Memory Usage	Uses the system call stack for each function call.	Uses a user-defined stack in memory
Execution flow	Function calls itself until the base condition is reached	Uses push and ~~loo~~ pop operations inside loops.
Stack overflow Risk	High for deep recursion because the call stack is limited	Lower risk because the stack size can be controlled or increased dynamically.

Conclusion :

* Recursion is simpler and easier to understand but may cause stack overflow for large inputs.

* Explicit stack provides better control over memory and is more suitable for large or deep problems.

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(2) A stack has a capacity of 5 elements. Explain what happens if the following operations are executed. Analytical problem solving.

Push(A), Push(B), Push(C), Push(D), Push(E), Push(F)

Identify the error and suggest two possible solutions.

Stack Capacity = 5

Operations :

Push(A) = ✓ Push(D) = ✓

Push(B) = ✓ Push(E) = ✓

Push(C) = ✓ Push(F) = ✗

⇒ After inserting A, B, C, D, E the stack becomes full.

⇒ When Push(F) is executed, there is no free space.

⇒ This causes a stack overflow error

Error :

⇒ Stack overflow - Attempting to push an element into a full stack.

Two Possible Solutions :

1. Increase the stack size so it can hold more elements.

2. Use a dynamic stack that grows as needed without a fixed size.